

GUREHMAT CHAHAL

Toronto, ON

647-400-8602 gurehmat9@gmail.com [linkedin.com/in/gurehmat](https://www.linkedin.com/in/gurehmat) github.com/Gurehmat gurehmat.dev

Education

McMaster University

Honours Bachelor of Applied Science, Computer Science (Co-op)

2025 – 2029

Hamilton, ON

Experience

AI Developer

2026

Workplace Edge

Remote

- Delivered an AI coaching chatbot for an enterprise client by building the full stack with **React 18**, **TypeScript**, **Express**, **Prisma**, and **Gemini 2.5 Flash**, deploying frontend on **Vercel** and API on **Railway** with **Turso** (**libSQL**).
- Improved response relevance by building an intent detection pipeline that classifies learner requests across 4 categories and injects matching lesson content into Gemini's system prompt.
- Prevented unsafe interactions by adding a content safety layer with PII stripping, keyword-based HR/crisis escalation detection, and input filtering that blocks flagged messages before persistence.

Founder & CTO

2024 – 2025

Advancing Future Innovators (AFI)

Ontario

- Founded a nonprofit delivering coding workshops to 30+ students per session, managing school partnerships, operations, and volunteer coordination across local schools.
- Designed and delivered structured programming curriculum covering Scratch, logic, and computational thinking.

Projects

The Record – Live AI Debate Referee | *React 18, TypeScript, Zustand, Gemini 2.5 Flash, ElevenLabs TTS*

2026

- Won **Best Use of ElevenLabs** at ConHacks 2026 by building solo a real-time debate referee in 36 hours that scores arguments on logic, evidence, and rhetoric, then delivers spoken rulings via **ElevenLabs TTS**.
- Integrated **Gemini 2.5 Flash** for real-time argument analysis and structured scoring, piping AI-generated rulings into **ElevenLabs** for text-to-speech delivery and managing async state (rounds, scores, audio) through **Zustand** stores.

PantryChef – AI-Driven Recipe Engine | *React 19, TypeScript, Gemini API, Zustand, Supabase, Tailwind CSS*

2026

- Enabled ingredient-to-recipe matching from fridge photos by integrating **Google Gemini** Vision API for scanning, recipe generation, and substitution suggestions, routed through **Supabase Edge Functions**.
- Achieved **O(P+R)** recipe search by designing a Set-based matching algorithm with confidence scoring (exact, partial, fuzzy), plus Cook Mode, expiry alerts, and response caching.
- Validated with **154 automated tests** (Vitest + RTL), strict **TypeScript**, CI/CD via **GitHub Actions**, Row-Level Security on all Postgres tables, and pre-commit hooks with ESLint + Prettier.

NASA Space Settlement Design Contest | *Grand Prize Winner (Healthy Living Design)*

2025

- Won the **Grand Prize** out of 9,000+ participants across 12 countries by authoring a 15-page technical proposal for a self-sustaining space settlement orbiting Phobos.
- Validated food production feasibility by designing a closed-loop aeroponic system for 800+ inhabitants and building an **Arduino**-based environmental monitoring prototype; presented at the ISDC conference.

Technical Skills

Languages: C, Python, JavaScript (ES6+), TypeScript, SQL, HTML/CSS, Haskell

Frameworks/Libraries: React 19, Zustand, Tailwind CSS, Supabase, Node.js, Vite, PostgreSQL

Tools: Git/GitHub, GitHub Actions, Vitest, ESLint, Prettier, Cursor, Linux, VS Code, Docker

Concepts: LLM Integration (Gemini), Prompt Engineering, REST APIs, CI/CD, Row-Level Security, State Management

Leadership / Extracurricular

SEED — MillHacks 2025

2025

Best UI Award | HTML, SCSS, JavaScript

- Won **Best UI** at MillHacks 2025 by building a single-page climate-tech site targeting coastal plastic pollution with modular SCSS architecture.

Activities: The Record (ConHacks 2026, Best Use of ElevenLabs), C.A.P.T.C.H.A. (Toronto Stupid Ideas Hackathon, solo), MEC Debate (2nd Place), Mac AI Club, Engineers Without Borders, Socratica